Building a Steel Bike Frame

From Prototype to Production

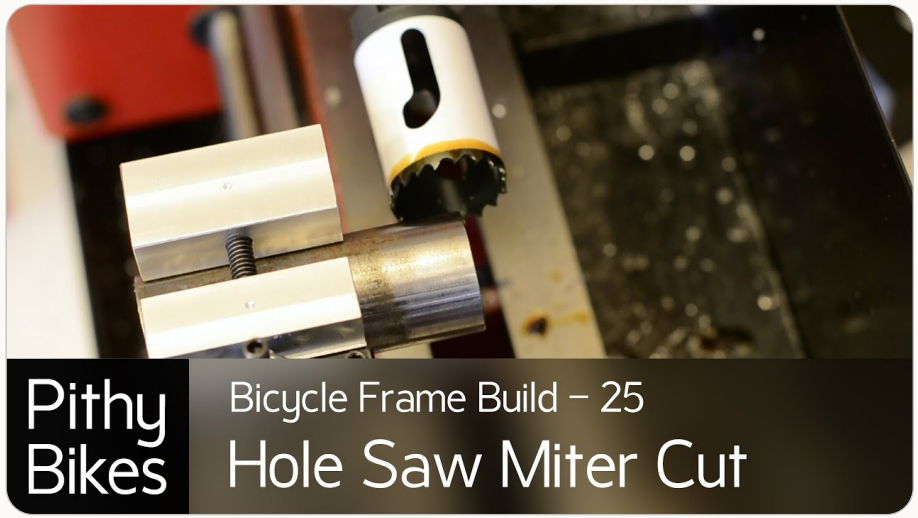
Manual Welding • Tube Preparation • Joining Methods
Structural Integrity • Repeatability • Manufacturing Reference

What Manual Frame Building Involves

- Start with proper butted steel tubes (4130 Chromoly, Reynolds, Columbus) and accurately miter/notch them for tight joints.
- Tube mitering can be done by hand with files or on a mill/hole-saw setup — the goal is tight, repeatable joints.
- A frame jig (commercial or DIY) holds the head tube, seat tube, BB shell, and stays in alignment during joining.
- Precision is everything: small gaps or misalignment lead to weak joints and handling issues.

Key Materials

4130 Chromoly • Reynolds 525/725 • Columbus tubing



Tube mitering with a hole saw fixture

Steel frame held in an Anvil frame jig

Joining Options

TIG Welding

TIG welding a bicycle frame joint

- Fuses tubes directly with filler rod
- Common for modern steel frames
- Requires good heat control on thin walls (~0.8 mm)
- Clean, precise results

Lugged Brazing

Brazed steel frame with lugs

- Uses steel lugs + silver/brass filler
- Tubes slip into lugs, heated, alloy wicks through
- Very approachable for small shops
- Classic aesthetic

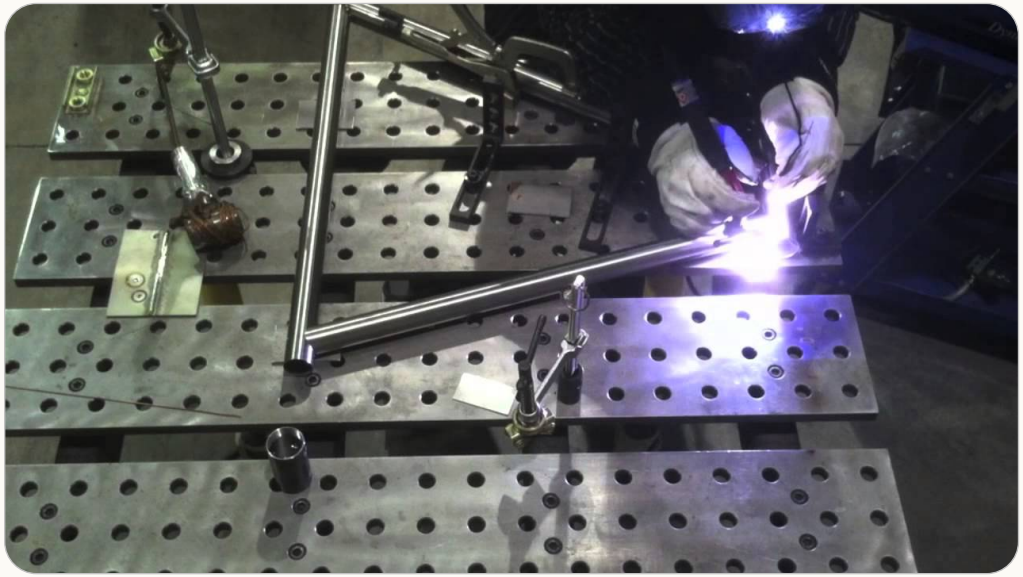
Fillet Brazing

Fillet brazed bottom bracket shell

- No lugs needed — smooth fillet at junctions
- Slower but forgiving process
- Widely used by custom builders
- Beautiful organic joints

Is It Structurally Sound?

When done correctly, joints are stronger than the thin tube itself.



Destructive testing a welded bicycle frame

What destructive tests show:

- Tube typically fails away from the joint, not at it
- Both brazed and TIG-welded joints exceed tube strength when properly executed

Key risks for DIY builders:

- Overheating thin butted sections
- Poor fit-up (gaps between tubes)
- Misalignment from skipping or improvising the jig

Tip: Practice on scrap tubes, cut joints open to inspect penetration before building a rideable frame.

Practical Considerations

What You Need

- ☑ **Frame design**
  Full-size drawing with geometry
- ☑ **Tubing set**
  BB shell, dropouts, head tube
- ☑ **Mitering setup**
  Mill/hole-saw or hand files
- ☑ **Jig/fixture**
  Commercial or DIY frame jig
- ☑ **Joining equipment**
  TIG welder or brazing torch
- ☑ **Inspection tools**
  Alignment gauges, calipers

Note on welding method:
MIG/stick can join steel, but TIG or brazing is strongly preferred for thin butted tubing — better control, less burn-through.

A first bicycle frame build from start to finish

Getting Started Faster

Many first-time builders take a short framebuilding class or follow detailed tutorials to shortcut the trial-and-error phase and get professional guidance.

Designing for Repeatability

What to Standardize

Frame Jig

Build or buy a jig with clear scales or hard stops to repeat key dimensions (BB drop, reach, chainstay length, head angle) without re-measuring.

Tube Mitering Fixtures

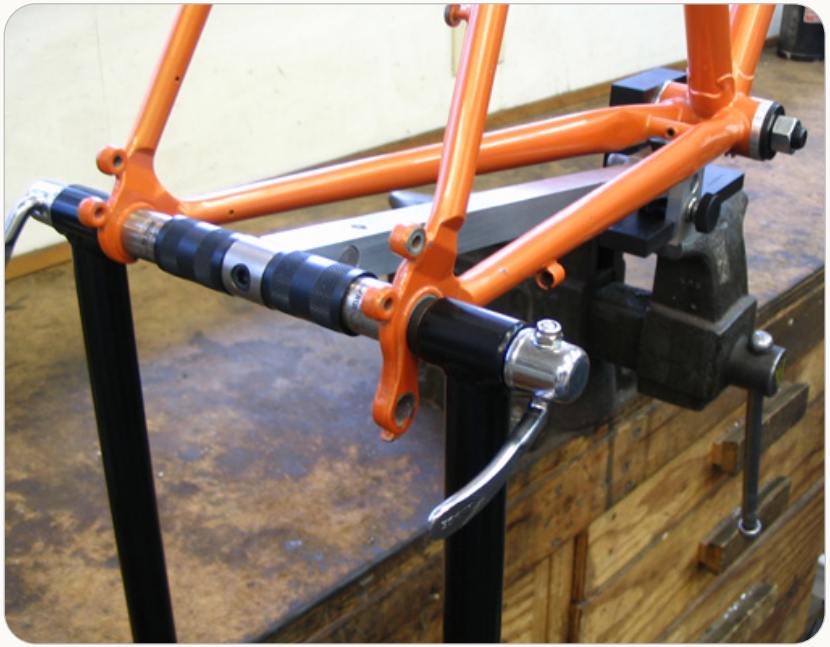
Use mill/holesaw fixtures or printed templates so miters land in the same place each time — making front triangles 'drop-in' parts.

Datum Strategy

Always reference from the same datums (BB center + head-tube centerline) for design, jig setup, and checks.

Good Starting Point: Steel Hardtail

- Fixed wheel size and travel (e.g., 29er, 130 mm fork)
- Standard BB type, head tube spec, and dropouts
- Only a few adjustable parameters per version



Precision frame alignment fixture by Stein Tools

From Prototype to Production

1 Build & Ride

Build a rideable steel prototype with intended geometry. Document everything with drawings and specs.

2 Iterate & Freeze

Ride and iterate until happy with handling, stiffness, and fit. Then freeze the design as reference.

3 CAD Package

Translate into clean 3D CAD + 2D drawings: tube specs, butting zones, joints, and tolerances.

What Changes in Overseas Production

- Different processes: hydroformed aluminum or carbon molds instead of hand-mitered steel
- Factory may suggest tweaks for manufacturability: simplifying junctions, adjusting wall thickness
- Pre-production cycle: factory sample, your testing, then approve or revise before first batch

3D CAD model of a mountain bike frame in Fusion 360

Making Your Prototype Factory-Friendly

1 Use Common Standards

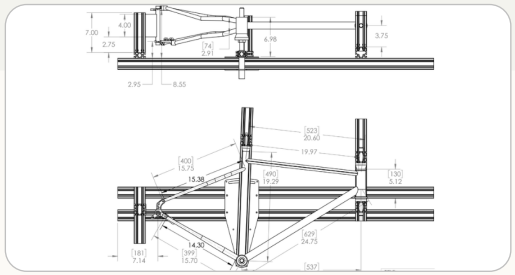
Standard head tube types, BB shells, dropout styles, and tire sizes are easier for suppliers to work with at scale.

2 Design with Clear Datums

Reference all dimensions from BB, rear axle, and head-tube centerline so the factory can match your numbers.

3 Document Actual vs. Design

Measure the final prototype frame (actual vs design) so you can tell if overseas samples are genuinely on-spec.



Sources

1. **Instructables** — Build a Bicycle Frame
   https://www.instructables.com/Build-a-Bicycle-Frame/
2. **YouTube** — Welding a Bike Frame on a StrongHand BuildPro Table
   https://www.youtube.com/watch?v=yT108iL-vMw
3. **Frame Builder Supply** — A Primer on Lugs
   https://framebuildersupply.com/pages/a-primer-on-lugs
4. **Bike Blog Ordie** — Brazing Practice and More
   https://www.bikeblogordie.com/2015/08/brazing-practice-and-more.html
5. **Metal-Guru** — TIG Welding for Bicycle Frame Builders
   https://metal-guru.com
6. **Reddit r/Welding** — Making a Steel Bicycle Frame
   https://www.reddit.com/r/Welding/comments/2c6w4o/making_a_steel_bicycle_frame/
7. **Pinhead Bike Locks** — How Bike Frames Are Built
   https://www.pinheadbikelocks.com/post/an-inside-look-at-how-bike-frames-are-built
8. **Haideli Bikes** — Carbon Fiber Frame Manufacturing
   https://haidelibikes.com/blog/carbon-fiber-bicycle-frame-manufacturing-process/
9. **Hont Carbon** — Guide to Bike Frame Design
   https://hontcarbon.com/blog/complete-guide-bike-frame-design-construction/
10. **Tideace Bike** — Custom Monocoque Carbon Frame
    https://tideacebike.com/blogs/buying-guide/custom-monocoque-carbon-frame
11. **MBR** — Framebuilders: The Full Story
    https://www.mbr.co.uk/news/product_news/framebuilders-the-full-story-280179